MTH631 - Real Analysis II (Quiz No 3)

Question # 6 of 10 (Start time: 04:14:25 PM, 01 January 2026)

Let $f(x,y) = \frac{xy}{x^2+y^2}$ then the limit of f along the line $y = -x$ as (x,y) approach $(0,0)$ is _____.

Select the correct option

○	undefined
●	$-1/2$
○	$1/2$
○	0

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Question # 5 of 10 (Start time: 04:14:12 PM, 01 January 2026)

A compact set in $\mathbb{R}^n$ is - - - - - - - - - - .

Select the correct option

○	open and bounded
○	open and unbounded
●	closed and bounded
○	closed and unbounded

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Question # 4 of 10 (Start time: 04:13:55 PM, 01 January 2026)

If $\mathbb{R}^n$ is connected, such that $\mathbb{R}^n = A \cup B$ with $\bar{A} \cap B = A \cap \bar{B} = \phi$, then - - - - - .

Select the correct option

○	$\bar{A} \subset A, \bar{B} \subset B$
◉	Either $A = \mathbb{R}^n, B = \phi$ or $B = \mathbb{R}^n, A = \phi$
○	A and B are both open and closed
○	All above can be concluded

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Question # 3 of 10 (Start time: 04:13:43 PM, 01 January 2026)

In $\mathbb{R}^n$, the interior S^0 of a non - empty set S is - - - - - - .

Select the correct option

○	open - disconnected
●	open - connected (region)

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Question # 2 of 10 (Start time: 04:13:22 PM, 01 January 2026)

If $\lim_{X \to X_0} f(X)$ exists, then it is _____.

Select the correct option

○	0
○	none of these
○	1
●	unique

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Quiz Start Time: 04:12 PM, 01 January 2026

Question # 1 of 10 (Start time: 04:12:59 PM, 01 January 2026) **Total Marks:** 1

In $\mathbb{R}^n$, if $S_1, S_2, S_3 \ldots$ are non - empty closed subsets such that $S_1 \supset S_2 \supset \cdots \supset S_r \supset \cdots$ and $\sup\{|X_r - Y_r| : X_r, Y_r \in S_r, r \geqslant 1\} \to 0$ as $r \to \infty$,then the order of set $\bigcap_{r=1}^{\infty} S_r = - - -$.

Select the correct option

Reload Math Equations

- infinite
- finite but more than one
- one
- zero

Click to Save Answer & Move to Next Question

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Question # 10 of 10 (Start time: 04:15:37 PM, 01 January 2026)

The limit of the function $f(x) = \frac{xy}{x^2+y^2}$ by letting (x, y)approach $(0, 0)$along the line $y = x$ is _____.

Select the correct option

○	Undefined
○	Finite
●	$\frac{1}{2}$
○	$-\frac{1}{2}$

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Question # 9 of 10 (Start time: 04:15:19 PM, 01 January 2026) Total Marks: 1

In $\mathbb{R}^n$, which of the following is true about $\phi = \{\}$ and $A = \{(a_1, a_2, \ldots, a_n), a_i \in \mathbb{R}, 1 \leqslant i \leqslant n, i \in \mathbb{N}\}$?

Select the correct option

Reload Math Equations

- Both are Disconnected
- Both are Connected
- ϕ is Disconnected and A is Connected
- ϕ is Connected and A is Disconnected

Click to Save Answer & Move to Next Question

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Question # 8 of 10 (Start time: 04:15:00 PM, 01 January 2026)

Set of isolated point(s) of the complement of set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $----$.

Select the correct option

○	$\{(x,y) \in \mathbb{R}^2 : x = y = \lvert n \rvert, n \in \mathbb{N}\}$
○	$\{(x,y) \in \mathbb{R}^2 : x = y = \lvert n \rvert, n \in \mathbb{N}\} \cup \{(0,0)\}$
◉	$\{(0,0)\}$
○	$\{(x,y) \in \mathbb{R}^2 : x = y = \lvert n \rvert, n \in \mathbb{N}\} \cap \{(0,0)\}$

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Question # 7 of 10 (Start time: 04:14:40 PM, 01 January 2026)

If $\phi \neq S \subseteq \mathbb{R}^n$, then the set S is bounded if - - - - - - - - .

Select the correct option

◉	$\sup\{\|X-Y\| : X, Y \in S\} < \infty$
○	$\inf\{\|X-Y\| : X, Y \in S\} < \infty$
○	$\inf\{\|X-Y\| : X, Y \in S\} = \infty$
○	$\sup\{\|X-Y\| : X, Y \in S\} = \infty$